

# Carlos A. Fernandes dos Santos

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## Summary

Data professional with a PhD in Physics and 10+ years of experience in using Data Science and Artificial Intelligence to solve challenging academic and business problems.

Experienced in architecting, designing, and productionizing advanced AI/ML solutions, including generative and agentic AI, to automate and transform operations at scale. Proven hands-on expertise and a strong track record in building, mentoring, and leading high-performing teams to deliver enterprise-grade solutions.

Strong skills in Python programming, mathematics, algorithms, MLOps, distributed computing, and cloud platforms.

## Experience

### JPMorgan Chase

*Applied AI/ML Director - Executive Director*

*Applied AI/ML Lead - Vice President*

**London, UK**

*Feb 2025 – Present*

*May 2023 – Jan 2025*

**Description:** Applied AI/ML Director within the Human Resources Data and Analytics team at JPMorgan Chase, focusing on leveraging artificial intelligence and machine learning to enhance employee experiences and drive business success. Main responsibilities include:

- Building and leading the EMEA HR AI/ML team to develop and deliver AI/ML solutions that support HR product areas, ensuring resource alignment and prioritization, while mentoring and inspiring a culture of innovation and continuous learning.
- Architecting, developing, and deploying advanced AI/ML solutions - including generative and agentic AI - to automate and optimize complex HR operational processes.
- Managing a portfolio of projects from inception to production, collaborating with stakeholders and cross-functional teams across the firm to deliver scalable, reliable and sustainable business solutions.
- Promoting reuse and partnership of AI/ML techniques and tools across the firm, and integrating vendor solutions where appropriate.
- Ensuring compliance with all applicable controls, policies, and procedures to maintain operational integrity.

### Microsoft

*Senior Specialised AI Cloud Solution Architect*

**London, UK**

*Jul 2021 – Apr 2023*

**Description:** Cloud Solution Architect, specialised in AI and Data Science on Azure, within the Microsoft Customer Experience and Success organisation. This role focuses on the strategy, design, and implementation of AI services and solutions for global customers. Main responsibilities include:

- Designing and building scalable AI/ML solutions that drive business value for customers. This is achieved by working with customers to develop Proofs of Concept and/or Minimum Viable Products, while ensuring a responsible use of AI.
- Supporting customers in the adoption of Machine Learning Operations (MLOps)
- Identifying and collaborating in the development of code and documentation assets that can be reused to further scale the adoption of Azure AI services.
- Liaising with Microsoft Engineering teams to remove key blockers and drive product improvements, based on Customer feedback.

### Accenture, Applied Intelligence

*Data Science Manager*

*Data Science Consultant*

**London, UK**

*Dec 2019 – Jun 2021*

*Oct 2018 – Nov 2019*

**Description:** Data Science Manager within the Applied Intelligence team, and part of the wider Strategy & Consulting practice. Key achievements include:

- Leading a global cross-functional team of Data Scientists, ML Engineers, Data Engineers and Frontend developers to develop digital solutions to improve and optimise processing plants for a mining organisation. These solutions, developed in Python and deployed in Azure, are to be scaled to multiple mining plants globally, and support the business goal of generating \$100M of value from data analytics in 2020.
- Managing, as the Data Science Lead, a team of 7 data scientists on a large project aiming to develop a solution that identifies product trends based on social media, for a large retail client. The solution is developed in Python, leverages NLP and ML techniques, and is deployed in GCP.
- Managing a team of 5 analysts and consultants. Key responsibilities include finding and securing chargeable projects, coaching and support with career development.

## Deloitte UK

Senior Consultant / Data Scientist  
Consultant / Data Scientist

London, UK

Sep 2017 – Oct 2018  
Feb 2016 – Aug 2017

**Description:** Data Scientist within the Risk Analytics team and part of the wider Risk Advisory Practice. The role consists in developing data science solutions to help organisations, within financial services, to gain market insights, minimise risk or drive process improvement and efficiency. Key achievements include:

- Developing a policy violation predictive model for a professional service firm, using Random Forests.
- Developing optimisation models for a Payment Protection Insurance (PPI) complaint operation, with a workforce of around 1000 people and approximately 100 000 complaints, for a large retail bank.
- Managing and developing a cognitive proof of concept, for a big four UK bank, to scrape and analyse large sets of news articles, in an automated way, aiming to support the credit underwriting process. This PoC was developed in Python and leveraged NLP and ML techniques.
- Developing of emotion, sentiment and topic models, leveraging NLP techniques in Python, to analyse audio files transcriptions, to automate the auditing process of several thousand hours of recorded phone calls, for a large insurance company.

## Istituto Nazionale di Fisica Nucleare - Sezione di Trieste

Postdoctoral Research Fellow

Trieste, Italy

Mar 2014 – Jan 2016

**Description:** Postdoctoral research fellowship in the field of experimental particle physics, focused on the research and development of gaseous photon detectors dedicated to the studies of the nucleon structure. The work performed envisaged the upgrade of a large area detector of a major experiment at a world leading physics laboratory in Geneva – CERN, where a significant part of R&D activities took place. The main activities included:

- Processing and analysis of comprehensive data sets to validate the R&D carried out. Data processing and analysis performed using C/C++, Matlab and others, in Windows and Unix environments.
- Monte Carlo Simulations to study physics interactions.
- Development of hardware.
- Management of data acquisition leading to a deep understanding of complex structured data.
- Supervision of undergraduate students in R&D activities.

## Education

### Ph.D. in Physics

University of Aveiro

Oct 2009 – Feb 2014

Aveiro, Portugal

**Description:** Research thesis under the title “Thick-Microstructures for MPGDs: Simulation and Experimental Studies”. The research work was carried out between Aveiro, CERN and INFN Trieste.

### M.Sc. in Engineering Physics

University of Aveiro

Oct 2008 – Jul 2009

Aveiro, Portugal

**Description:** Research dissertation under the title: “Simulation of Electron Avalanches in Radiation Gaseous Detectors”.

### B.Sc. in Engineering Physics

University of Aveiro

Sep 2004 – Oct 2008

Aveiro, Portugal

**Description:** Research project under the title: “Energy Weighting in a Single Photon Counting X-ray Detector based in a MHSP”.

## Personal Skills and Competencies

### Languages

- **English** ■■■■■ | Native Language
- **Portuguese** ■■■■■ | Native Language
- **Italian** ■■■■ | Level B2
- **French** ■■■ | Level B1

### Soft Skills

- Problem Solver
- Critical Thinker
- Storyteller
- Good Communicator
- Curious

### Technical Skills

- Python ■■■■■
- Matlab ■■■■
- R ■■■■
- C/C++ ■■■■
- Pandas / Numpy / Scipy ■■■■■
- Scikit-learn ■■■■■
- TensorFlow / Keras ■■■■
- Machine Learning / Deep Learning ■■■■
- MLOps ■■■■
- Generative AI ■■■■■
- Bash/Unix ■■■■
- SQL / NoSQL ■■■■
- Microsoft Azure ■■■■■
- Google Cloud Platform ■■■■
- Amazon Web Services ■■■■
- Quantum Computing ■■■■
- Excel ■■■■
- L<sup>A</sup>T<sub>E</sub>X ■■■■

## Certifications.....

- Azure AI Engineer Associate
- Azure Data Scientist Associate
- Azure AI Fundamentals
- Azure Fundamentals
- Google Cloud Platform - Professional Data Engineer
- Google Cloud Platform – Professional Cloud Architect
- Google Cloud Platform - Associate Cloud Engineer
- MIT Quantum Computing Fundamentals Online Program – a two-course certificate program to establish a foundation of knowledge for understanding the business implications of quantum computing.